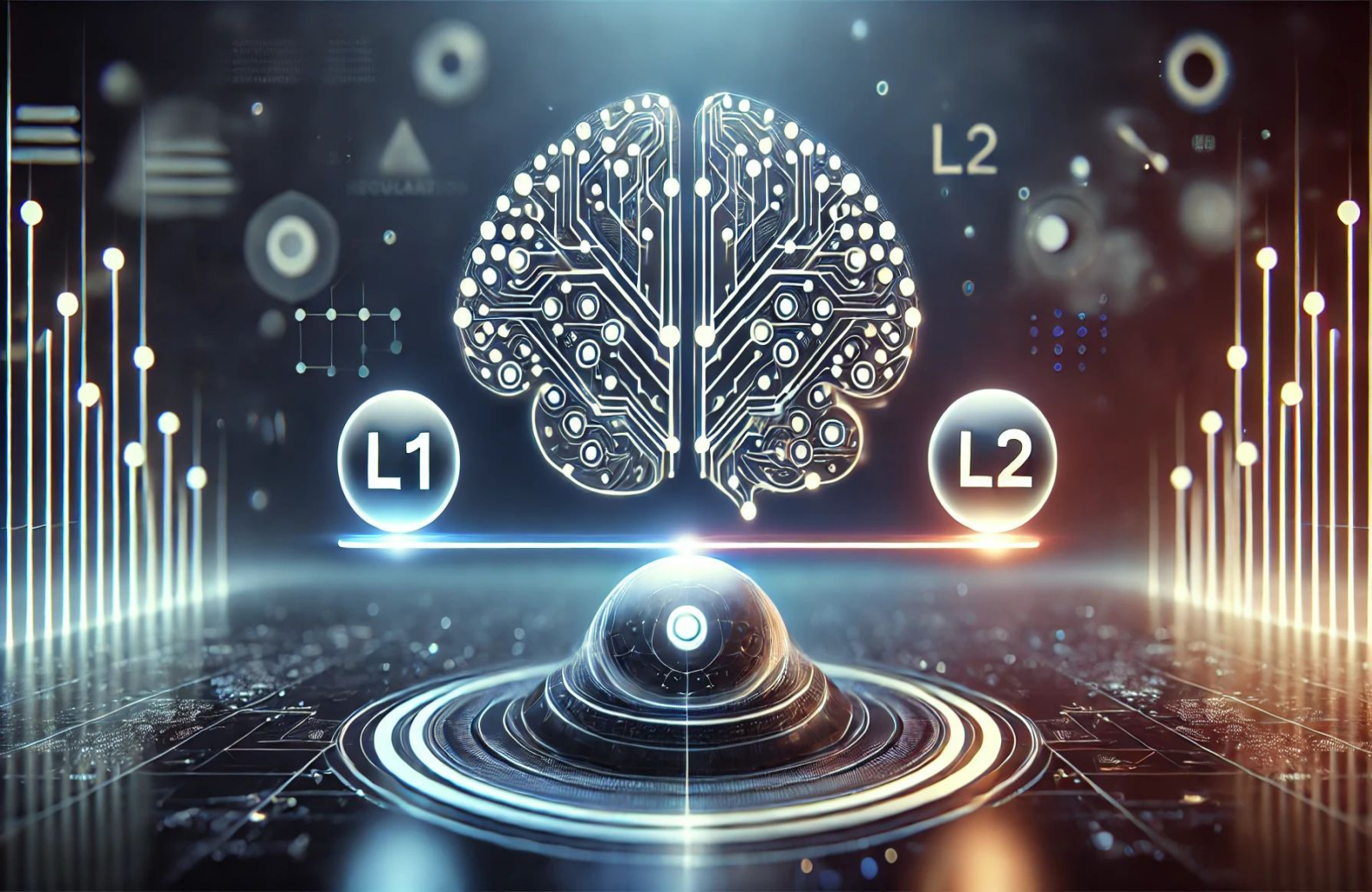




A Must-Know for ML Interviews!

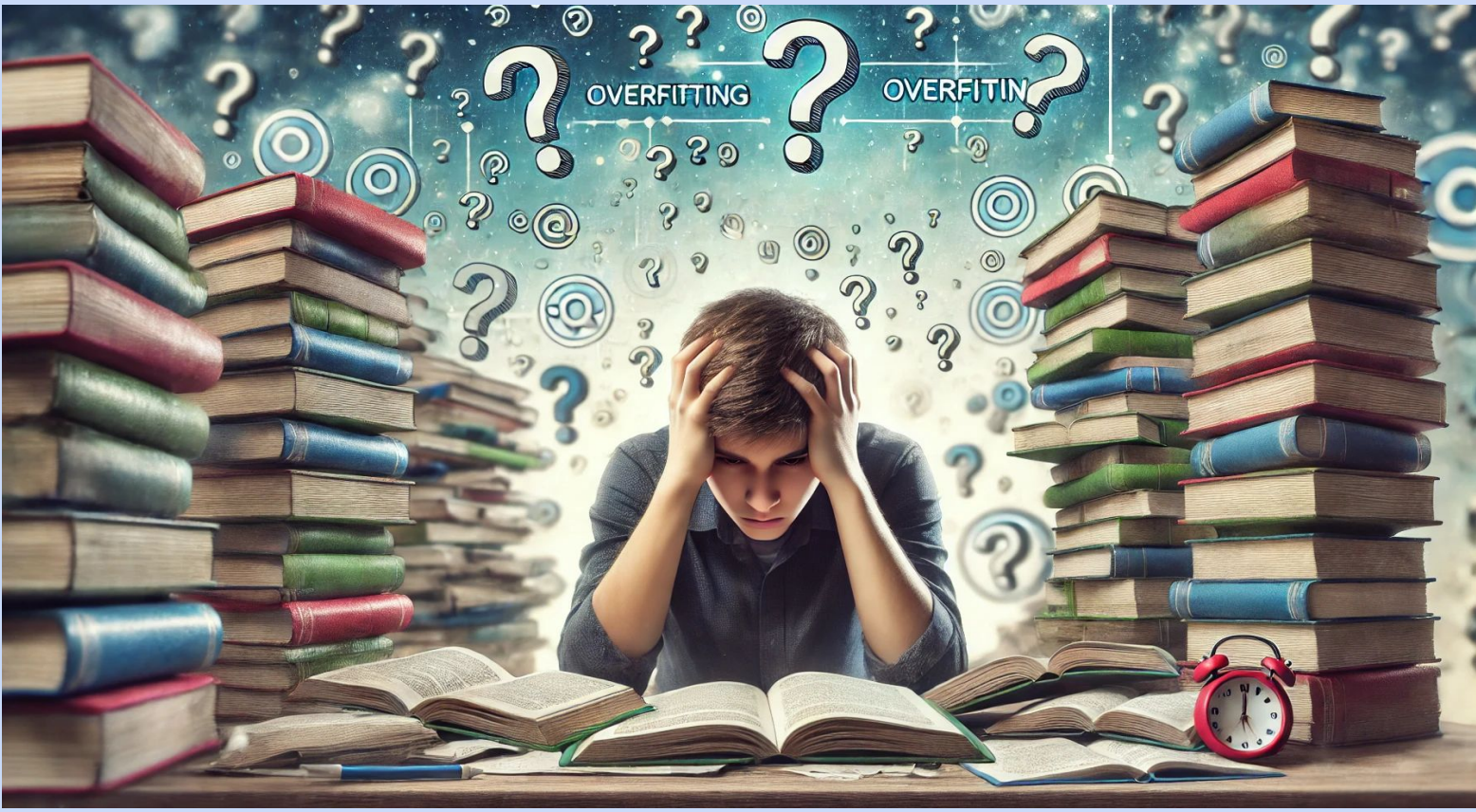
L1 vs. L2 Regularization – A Simple Guide



Sanjay N Kumar

Data scientist | AI ML Engineer | Statistician | Analytics Consultant

Why Do We Need Regularization?



🤔 Imagine this:

- You are studying for an exam. 📚
- You have **100 pages** to study, but only **10 pages** contain important concepts.
- What if you **memorized every word** instead of understanding key ideas? ❌
- You might **struggle in the exam** because you're overloaded with **unnecessary details**.

Why Do We Need Regularization?



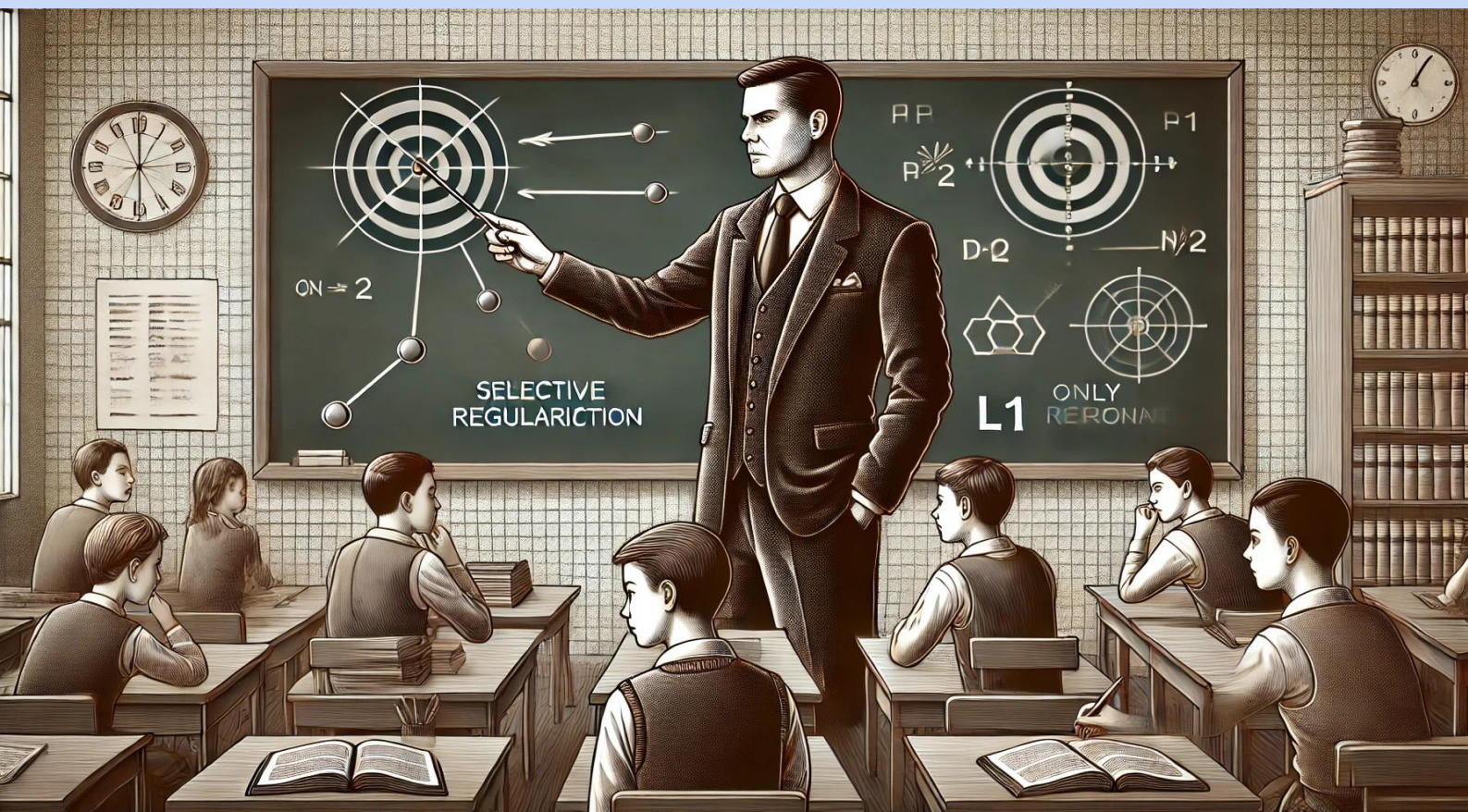
👉 This is **overfitting** in machine learning!

- The model memorizes every small detail instead of **learning the true pattern**.

✅ **Solution?**

L1 & L2 help AI focus on important information instead of memorizing everything.

Meet L1 Regularization (Lasso) 📌



- ◆ **L1 shrinks some features completely to zero** → meaning **unimportant features are removed**.
- ◆ It acts like a **strict teacher**, forcing the model to **focus only on the most important things**.

A top-down view of a packed suitcase and various travel items on a light gray surface. The open suitcase is filled with folded clothes, toiletries, and a passport. Surrounding it are shoes, a camera, a watch, sunglasses, and other personal items.

Example 1: Packing for a Trip

- You have a **small suitcase** and too many clothes.
- You **must pack only the most important items** (like your passport, phone, and wallet).
- Anything extra? **Gone!** 
-  **L1 removes unnecessary features, keeping only the essentials.**

Real-Life Examples for L1



📌 Example 2: Cooking a Dish 🍔

- Imagine you're making a simple **burger**.
- Do you really need **100 different spices**?
No!
- **L1 chooses only the key ingredients** (bread, butter, cheese) and ignores unnecessary ones.

Real-Life Examples for L1



Example 3: Study Notes

- You need to prepare for an exam.
- Instead of reading every page, you only focus on the most important points.
-  L1 picks only the essential features and removes the rest.

Math Behind L1

L1 **adds a penalty** to the sum of absolute values of weights:

$$Loss = Error + \lambda \sum |w|$$

(The more weights, the more penalty. **Only the most important weights survive!** 💡)

Meet L2 Regularization (Ridge) 📌



- ◆ L2 reduces all weights but never removes them completely.
- ◆ It ensures the model **spreads importance evenly** across all features instead of ignoring some.

Real-Life Examples for L2



Example 1: Tug of War

 Imagine a game of **tug of war**.

- If one team is much stronger, they will **dominate the game**.
- **L2 ensures no single player carries all the weight**—everyone **pulls together** with equal effort.

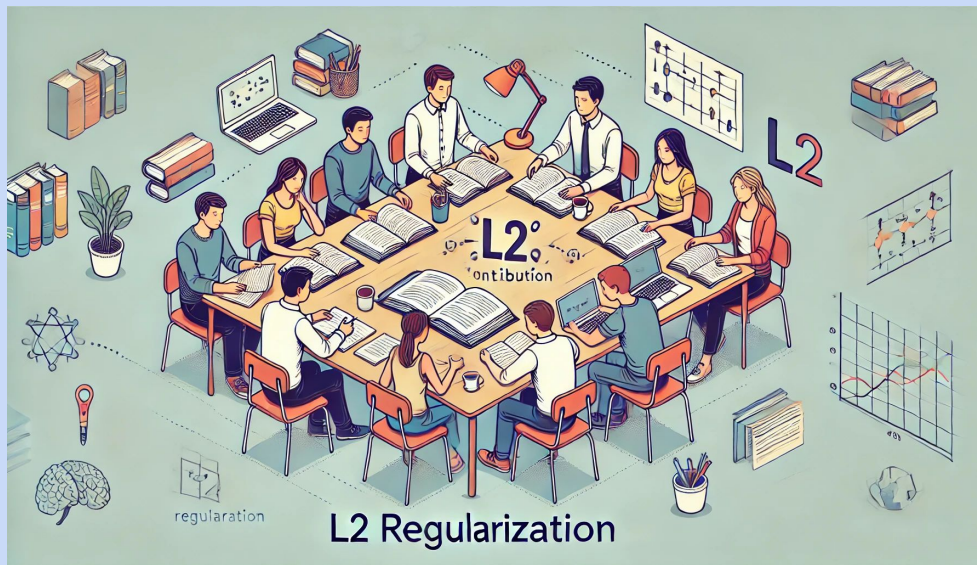
Real-Life Examples for L2



📌 Example 2: Cooking a Perfect Dish 🍲

- If you add too much **salt**, the taste is ruined!
- Instead, you **balance** all ingredients (spices, vegetables, and water).
- ☒ **L2 ensures every ingredient (feature) has some role, instead of removing any completely.**

Real-Life Examples for L2



📌 Example 3: Group Study Session 📖

📚 Imagine 5 friends studying together for an exam.

- **L1 (Lasso)** would say: *“Only the smartest student should study, and the rest can leave.”*
- **L2 (Ridge)** says: *“Everyone should contribute a little, so no one is overloaded.”*
-  **L2** spreads the learning evenly across all members, so the whole group improves together.

Real-Life Examples for L2



Example 4: Marathon Training

- If you **only train your legs** but ignore your breathing or endurance, you'll struggle in a real marathon.
-  **L2 ensures all aspects of fitness (speed, stamina, endurance) are balanced** instead of over-focusing on one.

Math Behind L2

L2 adds a **penalty** to the sum of squared weights:

$$Loss = Error + \lambda \sum w^2$$

(The penalty is applied to all features, but none are removed completely!)



Think of L2 like a fair coach! Instead of removing weak players, it trains everyone to contribute so the team performs as a whole!

When Should You Use L1 or L2?

✓ Use L1 if:

- ✓ You want to **remove unnecessary features** and keep only the most important ones.
- ✓ You need a **simpler model** with fewer inputs.

✓ Use L2 if:

- ✓ You want to **keep all features** but ensure none are too dominant.
- ✓ You want a **balanced, smooth** model where all variables contribute.

◆ **Use Both Together? – ElasticNet** combines both L1 & L2 for the best of both worlds!

Let's Connect It to Real-Life Scenarios!



◆ Scenario 1: Playing Football 🏈

- **L1:** Picks only the **best** players (star players like Messi, Ronaldo).
- **L2:** Ensures **everyone** gets a **chance to play**, making the team more balanced.

Let's Connect It to Real-Life Scenarios!



◆ Scenario 2: Cleaning Your Room 🏠

- **L1:** Throw away all unnecessary items and keep only what you need.
- **L2:** Tidy everything up, so all things have their place (nothing is thrown away completely).

Let's Connect It to Real-Life Scenarios!



◆ Scenario 3: Studying for an Exam 🎓

- **L1:** Focus only on **key chapters** and skip unnecessary details.
- **L2:** Study **everything** but with **equal effort**, even if some topics are less important.

L1 vs. L2 – What’s the Difference?

Feature	L1 Regularization (Lasso)	L2 Regularization (Ridge)
Main Idea	Selects only the most important features by making some weights exactly zero.	Reduces all weights but never makes them zero , ensuring every feature contributes.
Effect on Weights	Some feature weights become exactly zero (irrelevant features are removed).	All feature weights become smaller , but none are removed completely.
Best Used For	Feature selection – useful when you want a simpler model with only the most important variables.	Retaining all features but preventing any single feature from dominating the model.
Mathematical Formula	$(\text{Loss} = \text{Error} + \lambda \sum w)$	w^2
Sparsity?	✅ Yes, it makes the model sparse by removing unnecessary features.	❌ No, it keeps all features and only reduces their impact.
Computational Complexity	▲ Easier to interpret because many features are eliminated.	▼ Slightly more complex because all features are retained.
Overfitting Prevention	Helps prevent overfitting by removing unnecessary features.	Helps prevent overfitting by distributing feature influence.
Bias-Variance Tradeoff	Higher bias, lower variance (more stable but less flexible).	Lower bias, higher variance (more flexible but slightly less stable).
When to Use?	When you need a simpler model that only focuses on key variables.	When you want to keep all features but prevent overfitting by reducing their influence.

Final Thoughts

 **Think of L1 as an extreme filter – it keeps only the best of the best.**

 **Think of L2 as a balancing tool – it ensures all features contribute fairly.**

 **Still confused?** Imagine you are choosing your dream football team:

- **L1 picks only the best superstars.**
- **L2 makes sure everyone gets a fair chance to play.**

 **Next time you train a machine learning model, think about whether you need a strict coach (L1) or a fair teacher (L2)!**

The Power of Regularization

Master the Art of Balance in Machine Learning!

Let L1 sharpen your focus.

Let L2 bring harmony to your model.

Together, they turn complexity into clarity.



Choose wisely. Optimize boldly. Build smarter models!

Ready to take your models to the next level?

Let's connect!



Sanjay N Kumar

Data scientist | AI ML Engineer | Statistician | Analytics Consultant



<https://www.linkedin.com/in/sanjaytheanalyst360/>



sanjaytheanalyst360@gmail.com